

Project Design Phase-II Data Flow Diagram & User Stories

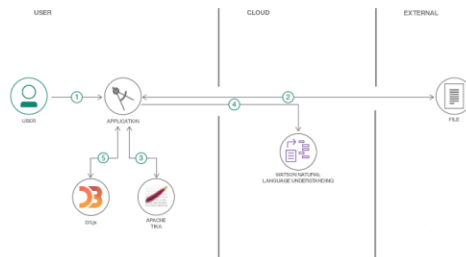
Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

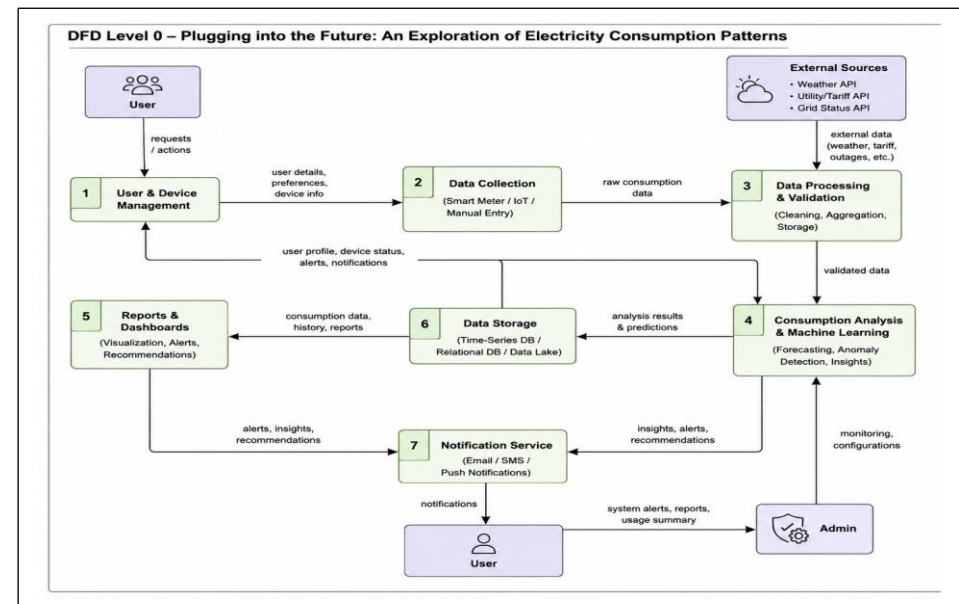
Example: (Simplified)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register by entering my name, email, phone number, and password.	I can successfully create an account and log in.	High	Sprint-1
	Registration	USN-2	As a user, I receive an email or OTP verification after registration.	My account is verified successfully.	High	Sprint-1
	Login	USN-3	As a user, I can log in using my registered email/mobile and password.	I can access my dashboard.	Low	Sprint-2
	Dashboard	USN-4	As a user, I can view my electricity consumption summary on the dashboard.	Dashboard displays daily, weekly, and monthly consumption.	Medium	Sprint-1
	Consumption Analysis	USN-5	As a user, I can view detailed electricity usage trends.	Graphs and charts display accurate consumption data.	High	Sprint-1
	Bill Prediction	USN-6	As a user, I can view my estimated electricity bill based on usage.	Estimated bill is generated using consumption data.	High	Sprint-2
	Alerts & Notifications	USN-7	As a user, I receive notifications when my electricity usage exceeds a threshold.	Alert is sent via app notification/email.	Medium	Sprint-2
	Energy Saving Tips	USN-8	As a user, I receive personalized energy-saving recommendations.	Relevant tips are displayed based on my usage.	Medium	Sprint-3
Customer (Web user)	Dashboard	USN-9	As a web user, I can view detailed analytics through an interactive dashboard.	Dashboard loads with charts and filters.	High	Sprint-2
	Data Comparison	USN-10	As a web user, I can compare electricity consumption across different time periods.	Comparison graphs are displayed correctly.	Medium	Sprint-2
	Profile Management	USN-11	As a web user, I can update my profile information.	Updated profile is saved successfully.	Low	Sprint-3
Customer Care Executive	Customer Support	USN-12	As a customer care executive, I can search users and view their electricity usage details.	Customer records are retrieved successfully.	High	Sprint-2
	Complaint Management	USN-13	As a customer care executive, I can register and track customer complaints.	Complaint status is updated successfully.	High	Sprint-2
	Report Generation	USN-14	As a customer care executive, I can generate customer usage reports.	Reports are generated accurately.	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	User Management	USN-15	As an administrator, I can manage user accounts.	Users can be added, updated, or removed.	High	Sprint-1
	Data Management	USN-16	As an administrator, I can upload and manage electricity consumption datasets.	Dataset uploads successfully and is validated.	High	Sprint-2
	Analytics Management	USN-17	As an administrator, I can monitor overall electricity consumption trends.	System-wide analytics dashboard is available.	High	Sprint-3
	Alert Configuration	USN-18	As an administrator, I can configure consumption thresholds for alerts.	Alert settings are saved and applied.	Medium	Sprint-3
	System Monitoring	USN-19	As an administrator, I can monitor application performance and user activities.	Logs and monitoring information are displayed correctly.	Medium	Sprint-3